

MATH 212

HOMEWORK DUE WEEK 7 MONDAY

Winter 2026

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Note: The solutions were converted from my notes to LaTeX using Google Gemini. I have reviewed and edited them for accuracy.

■ Question 7001.

□

Show that the best approximation (using the ℓ_∞ norm) of $f \in C^0(I)$ by a constant c is

$$c = \frac{1}{2} \left(\inf_{x \in I} f(x) + \sup_{x \in I} f(x) \right).$$

Solution. Let $f \in C^0(I)$ and let $c \in \mathbb{R}$. We wish to minimize the error in the ℓ_∞ norm:

$$E(c) = \|f - c\|_{\infty, I} = \sup_{x \in I} |f(x) - c|.$$

Let $M = \sup_{x \in I} f(x)$ and $m = \inf_{x \in I} f(x)$. For any $x \in I$, we have $m \leq f(x) \leq M$, which implies

$$f(x) - c \leq M - c \quad \text{and} \quad c - f(x) \leq c - m.$$

Thus, $|f(x) - c| \leq \max\{M - c, c - m\}$. It follows that

$$E(c) = \max\{M - c, c - m\}.$$

To minimize $E(c)$, we consider the functions $g_1(c) = M - c$ and $g_2(c) = c - m$. Note that g_1 is strictly decreasing and g_2 is strictly increasing. The minimum of their supremum occurs when $g_1(c) = g_2(c)$:

$$M - c = c - m \implies 2c = M + m \implies c^* = \frac{M + m}{2}.$$

The corresponding error is $E(c^*) = \frac{M - m}{2}$.

If $c > c^*$, then $c - m > c^* - m = \frac{M - m}{2}$, so $E(c) > E(c^*)$. Similarly, if $c < c^*$, then $M - c > M - c^* = \frac{M - m}{2}$, so $E(c) > E(c^*)$. Therefore, c^* is the unique minimizer. ■

■ **Question 7002.** □

In class, we proved that given $f \in C^0(I)$, there exists a $p_* \in \mathcal{P}_n$, called a best approximation to f , such that

$$\|f - p_*\|_{\infty, I} = \inf \{ \|f - p\|_{\infty, I} \mid p \in \mathcal{P}_n \}$$

The argument can be generalized to the following statement:

Theorem 0.0.1

Let $\mathcal{U}$ be a finite-dimensional subspace of a normed linear space X . Then, for every $f \in X$, there exists an element of best approximation $u^* \in \mathcal{U}$ such that

$$\|f - u^*\|_X = \inf \{ \|f - u\|_X \mid u \in \mathcal{U} \}$$

In this problem, we want to show that the requirement of $\mathcal{U}$ being **finite dimensional** cannot be dropped.

Let c_0 be the normed linear space of infinite sequences f such that

$$f = (\xi_1, \xi_2, \xi_3, \dots), \quad \xi_k \rightarrow 0, \quad \|f\| = \max_k |\xi_k|$$

and let $\mathcal{U}$ be the subspace

$$\mathcal{U} = \left\{ u = (u_1, u_2, \dots) \in c_0 \mid \sum_{k=1}^{\infty} \frac{u_k}{2^k} = 0 \right\}$$

Suppose $f = (\xi_1, \xi_2, \xi_3, \dots) \in c_0 \setminus \mathcal{U}_0$ and let $\lambda = \sum_{k=1}^{\infty} \frac{\xi_k}{2^k}$.

Show that

$$d(f, \mathcal{U}_0) := \inf_{u \in \mathcal{U}_0} \|f - u\| \leq |\lambda|$$

but $\|f - u\| > |\lambda|$ for all $u \in \mathcal{U}_0$.

In other words, the element of best approximation does not exist (i.e., the infimum is not achieved).

Hint: Construct a sequence u_n such that $\|f - u_n\| \rightarrow |\lambda|$.

Solution. **Part I.**

Following the hint, we construct a sequence of elements $u^{(n)} \in \mathcal{U}_0$ that approaches the desired distance. Define the sequence $u^{(n)}$ by:

$$u^{(1)} = f - \frac{2}{1}\lambda(1, 0, 0, \dots), \quad u^{(2)} = f - \frac{4}{3}\lambda(1, 1, 0, 0, \dots), \quad u^{(3)} = f - \frac{8}{7}\lambda(1, 1, 1, 0, 0, \dots), \quad \text{etc.,}$$

and in general,

$$u^{(n)} = f - \left(1 - \frac{1}{2^n}\right)^{-1} \underbrace{\lambda(1, 1, \dots, 1, 0, 0, \dots)}_{n \text{ times}}.$$

Explicitly, the k -th term of $u^{(n)}$ is given by:

$$u_k^{(n)} = \begin{cases} \xi_k - (1 - 2^{-n})^{-1} \lambda & \text{for } 1 \leq k \leq n, \\ \xi_k & \text{for } k > n. \end{cases}$$

First, we verify that $u^{(n)} \in \mathcal{U}_0$ for each n . Evaluating the defining series for the subspace:

$$\begin{aligned} \sum_{k=1}^{\infty} \frac{u_k^{(n)}}{2^k} &= \sum_{k=1}^{\infty} \frac{\xi_k}{2^k} - (1 - 2^{-n})^{-1} \lambda \sum_{k=1}^n \frac{1}{2^k} \\ &= \lambda - (1 - 2^{-n})^{-1} \lambda (1 - 2^{-n}) \\ &= \lambda - \lambda = 0. \end{aligned}$$

Thus, $u^{(n)} \in \mathcal{U}_0$.

Next, we compute the distance $\|f - u^{(n)}\|$. The difference sequence is:

$$f - u^{(n)} = (1 - 2^{-n})^{-1} \lambda \underbrace{(1, 1, \dots, 1, 0, 0, \dots)}_{n \text{ times}}.$$

The sup-norm of this difference in c_0 is simply the maximum magnitude of its components:

$$\|f - u^{(n)}\| = \left(1 - \frac{1}{2^n}\right)^{-1} |\lambda|.$$

As $n \rightarrow \infty$, the factor $(1 - 2^{-n})^{-1}$ strictly decreases to 1. Therefore, $\|f - u^{(n)}\| \searrow |\lambda|$.

Part II.

Let $u \in \mathcal{U}_0$. By the definition of the subspace $\mathcal{U}_0$, we have $\sum_{k=1}^{\infty} \frac{u_k}{2^k} = 0$. Using the definition of λ , we can express it as:

$$\lambda = \sum_{k=1}^{\infty} \frac{\xi_k}{2^k} = \sum_{k=1}^{\infty} \frac{\xi_k}{2^k} - \sum_{k=1}^{\infty} \frac{u_k}{2^k} = \sum_{k=1}^{\infty} \frac{\xi_k - u_k}{2^k}.$$

Applying the triangle inequality, we obtain:

$$|\lambda| = \left| \sum_{k=1}^{\infty} \frac{\xi_k - u_k}{2^k} \right| \leq \sum_{k=1}^{\infty} \frac{|\xi_k - u_k|}{2^k}.$$

By the definition of the sup-norm in c_0 , $|\xi_k - u_k| \leq \|f - u\|$ for all $k \geq 1$. Consequently:

$$|\lambda| \leq \sum_{k=1}^{\infty} \frac{\|f - u\|}{2^k} = \|f - u\| \sum_{k=1}^{\infty} \frac{1}{2^k} = \|f - u\|(1) = \|f - u\|.$$

Suppose, for the sake of contradiction, that equality holds, meaning $\|f - u\| = |\lambda|$. For the equality to hold in the series bound above, we must have exactly $\xi_k - u_k = \lambda$ for all $k \geq 1$. (Any deviation in magnitude or sign would result in a strictly smaller sum, as the geometric series strictly weights the terms).

If $\xi_k - u_k = \lambda$ for all k , then $u_k = \xi_k - \lambda$. Since $f \in c_0$, we know $\xi_k \rightarrow 0$ as $k \rightarrow \infty$. This implies that $u_k \rightarrow -\lambda$. However, since $u \in \mathcal{U}_0 \subset c_0$, we must have $u_k \rightarrow 0$, which forces $\lambda = 0$.

If $\lambda = 0$, then $\sum_{k=1}^{\infty} \frac{\xi_k}{2^k} = 0$, meaning $f \in \mathcal{U}_0$. This contradicts the initial hypothesis that $f \in c_0 \setminus \mathcal{U}_0$. Thus, equality cannot hold, and we conclude that $\|f - u\| > |\lambda|$ for all $u \in \mathcal{U}_0$.

Since $d(f, \mathcal{U}_0) = \inf_{u \in \mathcal{U}_0} \|f - u\|$, the existence of the sequence in part I implies $d(f, \mathcal{U}_0) \leq |\lambda|$. Combined with Part II, we conclude that the distance is exactly $|\lambda|$ but the infimum is never achieved by any element strictly inside the subspace $\mathcal{U}_0$. In other words, the element of best approximation does not exist. ■

■ **Question 7003.**

An operator is a linear transformation from one vector space to another vector space.

Definition 0.0.2

Given norms $\|\cdot\|_V$ and $\|\cdot\|_W$ on vector spaces V and W , and an operator $A : V \rightarrow W$, define

$$\|A\|_{V \rightarrow W} = \sup_{0 \neq \vec{v} \in V} \frac{\|A\vec{v}\|_W}{\|\vec{v}\|_V}.$$

(a) Prove that

$$\|AB\|_{U \rightarrow W} \leq \|A\|_{V \rightarrow W} \cdot \|B\|_{U \rightarrow V}$$

Note: This is the triangle inequality requirement for showing that the definition forms a norm on the $\mathcal{L}(V, W)$, the vector space of all linear operators from V to W .

(b) Given a linear operator $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ (which is essentially a matrix), prove that the **induced** norm $\|A\|_p$ defined as $\|A\|_{\ell_p \rightarrow \ell_p}$ satisfies

$$\|A\|_p = \max_{\|\vec{x}\|_p=1} \|A\vec{x}\|_p.$$

Solution. (a) Let $B : U \rightarrow V$ and $A : V \rightarrow W$ be bounded linear operators. We wish to bound the norm of the composition $AB : U \rightarrow W$.

By the definition of the operator norm, for any $\vec{v} \in V$, we have the inequality $\|A\vec{v}\|_W \leq \|A\|_{V \rightarrow W} \|\vec{v}\|_V$. Let $\vec{u} \in U$ be an arbitrary non-zero vector. Applying the above inequality with $\vec{v} = B\vec{u}$, we obtain:

$$\|(AB)\vec{u}\|_W = \|A(B\vec{u})\|_W \leq \|A\|_{V \rightarrow W} \|B\vec{u}\|_V.$$

Similarly, applying the definition of the operator norm for B , we have $\|B\vec{u}\|_V \leq \|B\|_{U \rightarrow V} \|\vec{u}\|_U$. Substituting this into our previous inequality yields:

$$\|(AB)\vec{u}\|_W \leq \|A\|_{V \rightarrow W} (\|B\|_{U \rightarrow V} \|\vec{u}\|_U).$$

Dividing both sides by $\|\vec{u}\|_U$ (since $\vec{u} \neq 0$), we get:

$$\frac{\|(AB)\vec{u}\|_W}{\|\vec{u}\|_U} \leq \|A\|_{V \rightarrow W} \cdot \|B\|_{U \rightarrow V}.$$

Because this holds for all $\vec{u} \neq 0$ in U , it must hold for the supremum over all such $\vec{u}$. Thus, by definition:

$$\|AB\|_{U \rightarrow W} = \sup_{0 \neq \vec{u} \in U} \frac{\|(AB)\vec{u}\|_W}{\|\vec{u}\|_U} \leq \|A\|_{V \rightarrow W} \cdot \|B\|_{U \rightarrow V}.$$

(b) By definition, the induced p -norm of the operator $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is given by:

$$\|A\|_p = \sup_{\vec{x} \neq 0} \frac{\|A\vec{x}\|_p}{\|\vec{x}\|_p}.$$

For any $\vec{x} \neq 0$, let $\vec{u} = \frac{\vec{x}}{\|\vec{x}\|_p}$. By the absolute homogeneity of vector norms, $\|\vec{u}\|_p = \left\| \frac{\vec{x}}{\|\vec{x}\|_p} \right\|_p = \frac{\|\vec{x}\|_p}{\|\vec{x}\|_p} = 1$.

By the linearity of A and the homogeneity of the norm, we can rewrite the quotient as:

$$\frac{\|A\vec{x}\|_p}{\|\vec{x}\|_p} = \left\| A \left(\frac{\vec{x}}{\|\vec{x}\|_p} \right) \right\|_p = \|A\vec{u}\|_p.$$

This shows that taking the supremum over all non-zero vectors $\vec{x} \in \mathbb{R}^n$ is entirely equivalent to taking the supremum over all vectors $\vec{u}$ on the unit sphere in $\mathbb{R}^n$ (where $\|\vec{u}\|_p = 1$). Therefore:

$$\|A\|_p = \sup_{\|\vec{u}\|_p=1} \|A\vec{u}\|_p.$$

We now establish that this supremum is actually a maximum. The unit sphere $S = \{\vec{u} \in \mathbb{R}^n \mid \|\vec{u}\|_p = 1\}$ is a closed and bounded subset of the finite-dimensional space $\mathbb{R}^n$, which implies S is compact by the Heine-Borel theorem.

Furthermore, any linear transformation $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous, and any vector norm $\|\cdot\|_p$ is a continuous function. Thus, the composition $f(\vec{u}) = \|A\vec{u}\|_p$ is a continuous function defined on the compact set S. By the Extreme Value Theorem, f must attain its maximum value on S. Consequently, the supremum is achieved, and we may write:

$$\|A\|_p = \max_{\|\vec{x}\|_p=1} \|A\vec{x}\|_p.$$

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■ **Question 7004.**

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Note: The two problems are unrelated. I just didn't want to increase the total point count/number of questions. Also, both of these have hints in the class notes.

- (a) Suppose $\{\varphi_n\}$ is an orthogonal family of polynomials with respect to some weight function $w(x)$. Show that the degree n polynomial φ_n has n **distinct** real roots.
- (b) Check that although Simpson's rule is based on quadratic interpolants, it is exact for all cubic polynomials.

Solution. (a) Let $\{\varphi_n\}_{n=0}^{\infty}$ be a sequence of orthogonal polynomials on the interval (a, b) with respect to the weight function $w(x) > 0$. We assume $\deg(\varphi_n) = n$. Let $x_1, x_2, \dots, x_k$ be the distinct roots of $\varphi_n(x)$ that lie strictly inside the interval (a, b) and at which $\varphi_n(x)$ changes sign. Note that roots with even multiplicity do not cause a sign change.

Since a polynomial of degree n has at most n roots, we know $k \leq n$. Suppose for contradiction that $k < n$. Construct the polynomial

$$q(x) = \prod_{j=1}^k (x - x_j).$$

This polynomial has degree $k < n$, and thus $q \in \mathcal{P}_{n-1}$. By the orthogonality property, φ_n is orthogonal to all polynomials of degree strictly less than n , which implies

$$\langle \varphi_n, q \rangle = \int_a^b \varphi_n(x) q(x) w(x) dx = 0.$$

However, consider the product $\varphi_n(x)q(x)$. The roots of φ_n in (a, b) where the sign changes are exactly $x_1, \dots, x_k$. The polynomial $q(x)$ also changes sign exactly at these points. Therefore, the product $\varphi_n(x)q(x)$ does not change sign anywhere on the interval (a, b) .

Because $\varphi_n(x)q(x)$ maintains a constant sign (either non-negative or non-positive everywhere on (a, b)) and is not identically zero, we have

$$\int_a^b \varphi_n(x) q(x) w(x) dx \neq 0.$$

This contradicts the orthogonality condition. Therefore, our assumption that $k < n$ must be false. It follows that $k = n$, meaning φ_n has exactly n distinct real roots in (a, b) .

- (b) Simpson's rule on the interval $[-h, h]$ approximates the integral $\int_{-h}^h f(x) dx$ using the quadrature formula:

$$S(f) = \frac{h}{3} [f(-h) + 4f(0) + f(h)].$$

Since both integration and the quadrature rule are linear operators, it suffices to check exactness on a basis for $\mathcal{P}_3$, such as $\{1, x, x^2, x^3\}$. By construction, as it integrates the quadratic interpolant

exactly, $S(f)$ is exact for polynomials of degree up to 2. It remains only to verify exactness for $f(x) = x^3$.

The exact integral of x^3 over the symmetric interval $[-h, h]$ is:

$$\int_{-h}^h x^3 dx = \left[\frac{x^4}{4} \right]_{-h}^h = \frac{h^4}{4} - \frac{(-h)^4}{4} = 0.$$

Evaluating Simpson's rule approximation for x^3 yields:

$$S(x^3) = \frac{h}{3} [(-h)^3 + 4(0)^3 + (h)^3] = \frac{h}{3} [-h^3 + 0 + h^3] = 0.$$

Since the exact integral matches the quadrature approximation for x^3 , Simpson's rule is exact for the standard basis of cubic polynomials, and thus exact for all $p \in \mathcal{P}_3$.



■ **Question 7005.**

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Find the exact formula for the Gaussian quadrature with two nodes $Q_{G,1}$ with weight function $w(x) = 1$ on the interval $[-1, 1]$. We are going to do it in two ways:

(a) Write down the four equations:

$$w_0 f(x_0) + w_1 f(x_1) = \int_{-1}^1 f(x) dx$$

when $f(x) = 1, x, x^2, x^3$ and solve for the unknown nodes and weights.

(b) Use the roots of the second Legendre Polynomial φ_2 to construct the Lagrange basis and find the weights from the definition of $Q_{G,1}$.

Solution. (a) We require the quadrature rule $w_0 f(x_0) + w_1 f(x_1)$ to be exact for polynomials of degree up to $2n + 1 = 2(1) + 1 = 3$. Substituting $f(x) = 1, x, x^2, x^3$ yields the following system of four nonlinear equations:

$$w_0 \cdot 1 + w_1 \cdot 1 = \int_{-1}^1 1 dx = 2 \quad (1)$$

$$w_0 x_0 + w_1 x_1 = \int_{-1}^1 x dx = 0 \quad (2)$$

$$w_0 x_0^2 + w_1 x_1^2 = \int_{-1}^1 x^2 dx = \frac{2}{3} \quad (3)$$

$$w_0 x_0^3 + w_1 x_1^3 = \int_{-1}^1 x^3 dx = 0 \quad (4)$$

From equation (2), we have $w_1 x_1 = -w_0 x_0$.

Substituting this into equation (4) gives:

$$w_0 x_0^3 + (-w_0 x_0) x_1^2 = 0 \implies w_0 x_0 (x_0^2 - x_1^2) = 0.$$

Since $w_0 \neq 0$ (otherwise we would only have a one-node rule) and $x_0 \neq x_1$ (nodes must be distinct), we must have $x_0 \neq 0$. Thus, $x_0^2 - x_1^2 = 0$, which implies $x_1 = -x_0$.

Returning to equation (2), we now have $w_0 x_0 + w_1 (-x_0) = 0$, which gives $x_0 (w_0 - w_1) = 0$. Since $x_0 \neq 0$, this implies $w_0 = w_1$.

Using $w_0 = w_1$ in equation (1) gives $2w_0 = 2$, so $w_0 = 1$ and $w_1 = 1$.

Finally, using $w_0 = w_1 = 1$ and $x_1 = -x_0$ in equation (3) gives:

$$(1)x_0^2 + (1)(-x_0)^2 = \frac{2}{3} \implies 2x_0^2 = \frac{2}{3} \implies x_0^2 = \frac{1}{3} \implies x_0 = \pm \frac{1}{\sqrt{3}}.$$

Conventionally, we order the nodes such that $x_0 < x_1$. Therefore, the nodes are $x_0 = -\frac{1}{\sqrt{3}}$ and $x_1 = \frac{1}{\sqrt{3}}$, with weights $w_0 = 1$ and $w_1 = 1$.

The exact formula is:

$$\int_{-1}^1 f(x) dx \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right).$$

- (b) The fundamental theorem of Gaussian quadrature states that the optimal nodes for the interval $[-1, 1]$ with weight function $w(x) = 1$ are the roots of the n -th degree Legendre polynomial, $P_n(x)$. For a two-node rule ($n = 2$), we need the roots of $P_2(x)$.

The Legendre polynomials are orthogonal on $[-1, 1]$. Using the standard recurrence or Gram-Schmidt orthogonalization on $\{1, x, x^2\}$, the monic orthogonal polynomial of degree 2 is:

$$\varphi_2(x) = x^2 - \frac{1}{3}.$$

Setting $\varphi_2(x) = 0$ immediately yields the nodes:

$$x_0 = -\frac{1}{\sqrt{3}}, \quad x_1 = \frac{1}{\sqrt{3}}.$$

The weights w_0 and w_1 are found by integrating the corresponding Lagrange basis polynomials over $[-1, 1]$:

$$\begin{aligned} w_0 &= \int_{-1}^1 \frac{x - x_1}{x_0 - x_1} dx = \int_{-1}^1 \frac{x - 1/\sqrt{3}}{-2/\sqrt{3}} dx = -\frac{\sqrt{3}}{2} \left[\frac{x^2}{2} - \frac{x}{\sqrt{3}} \right]_{-1}^1 \\ &= -\frac{\sqrt{3}}{2} \left(\left(\frac{1}{2} - \frac{1}{\sqrt{3}} \right) - \left(\frac{1}{2} + \frac{1}{\sqrt{3}} \right) \right) = -\frac{\sqrt{3}}{2} \left(-\frac{2}{\sqrt{3}} \right) = 1. \end{aligned}$$

Similarly,

$$\begin{aligned} w_1 &= \int_{-1}^1 \frac{x - x_0}{x_1 - x_0} dx = \int_{-1}^1 \frac{x + 1/\sqrt{3}}{2/\sqrt{3}} dx = \frac{\sqrt{3}}{2} \left[\frac{x^2}{2} + \frac{x}{\sqrt{3}} \right]_{-1}^1 \\ &= \frac{\sqrt{3}}{2} \left(\left(\frac{1}{2} + \frac{1}{\sqrt{3}} \right) - \left(\frac{1}{2} - \frac{1}{\sqrt{3}} \right) \right) = \frac{\sqrt{3}}{2} \left(\frac{2}{\sqrt{3}} \right) = 1. \end{aligned}$$

This confirms the nodes $x_0 = -1/\sqrt{3}, x_1 = 1/\sqrt{3}$ and weights $w_0 = 1, w_1 = 1$. The resulting formula is identical to the one derived in part (a).

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